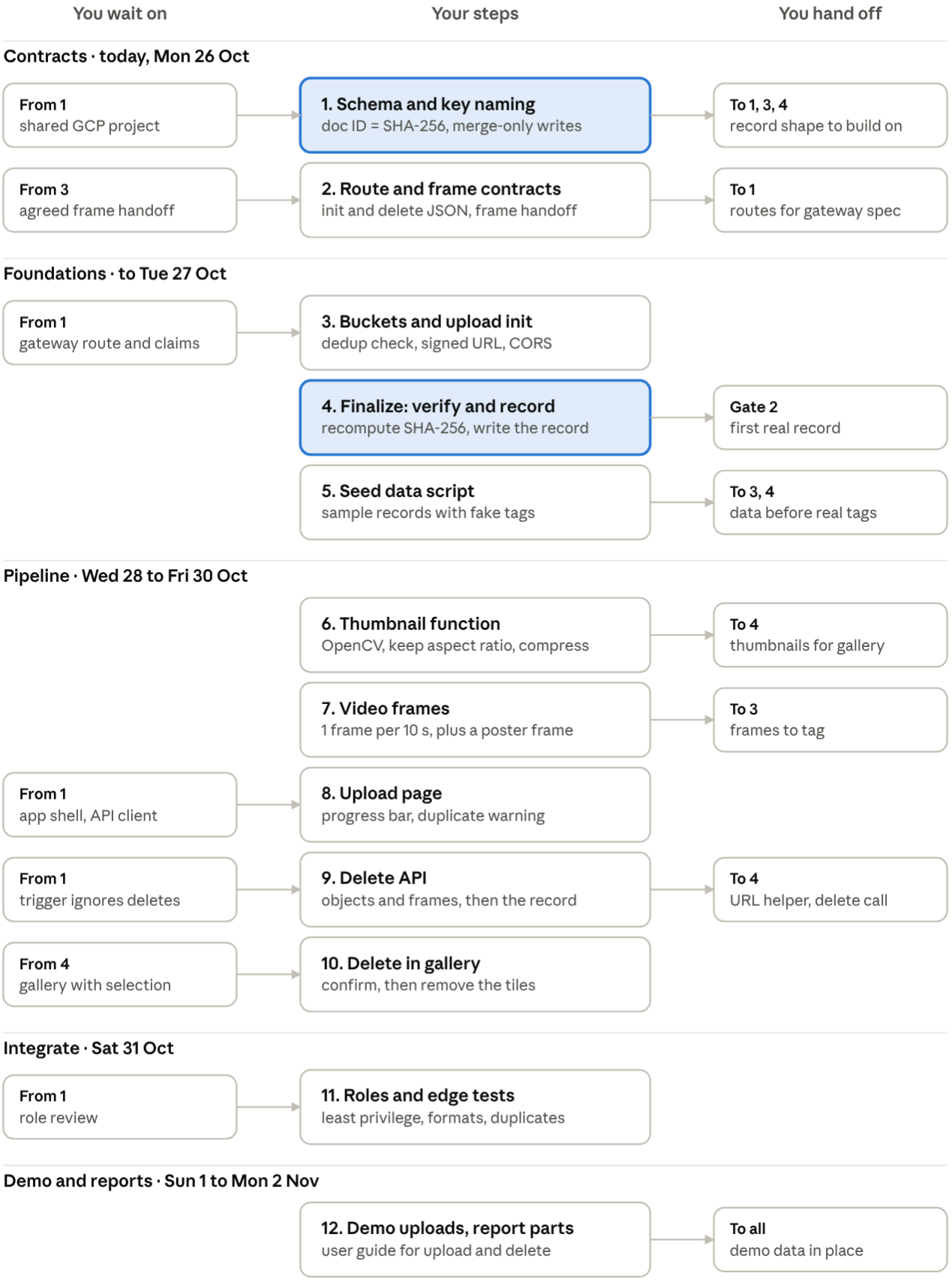
# Member 2 roadmap: ingestion and storage

Finish the schema and contracts today, because members 1, 3 and 4 all build on them. Then get a real upload writing its Firestore record by Tuesday. That is Gate 2, and member 3 cannot connect tagging to uploads until it passes; everything after it runs alongside your teammates.

## Finish the schema today and a real upload by Tuesday; the team builds on both

Your 12 steps in order; left boxes are what you wait on, right boxes are what teammates wait on




member 2 roadmap · 12 steps, 25 Oct to 2 Nov

The two highlighted steps are the ones teammates are blocked on. Everything on the right is a promise to someone else, so tell them when each one lands.

## Step details

| Step | Done when | Watch out for |
|---|---|---|
| 1. Schema and key naming | A schema file in the repo lists every field, its type and which function writes it, and the team has read it | Tags stay a species-to-count map; store bucket and key, never presigned URLs |
| 2. Route and frame contracts | Request and response JSON for `/uploads/init` and `/files/delete` is with member 1; the frame handoff is agreed with member 3 | Decide whether frames pass in memory or through the derived bucket before either of you writes code |
| 3. Buckets and upload init | The browser gets a signed PUT URL, or a duplicate response for a known hash | Signing from Cloud Run needs the Service Account Token Creator role on the function's own service account; bucket CORS must allow PUT from the app's origin and localhost; accept only formats OpenCV reads (HEIC is not one) |
| 4. Finalize: verify and record | Uploading a file creates its Firestore record within seconds | Recompute the SHA-256, since the browser's claim can't be trusted, and delete the object on a mismatch; carry the uploader's ID from init as object metadata |
| 5. Seed data script | One command loads about 20 sample records with varied tags | Use the real schema and naming so teammates' code works unchanged on real data; mark seeds so they are easy to wipe |
| 6. Thumbnail function | Every image gets `thumbs/<sha256>.jpg` and its record shows the key | Put thumbnails in a second bucket: Cloud Storage triggers fire for every object in a bucket, so a thumbnail in the media bucket re-triggers the function |

| Step | Done when | Watch out for |
|---|---|---|
| 7. Video frames | A 60 s clip gives frames at 0, 10, 20, 30, 40 and 50 s, plus a poster thumbnail | Seek by time instead of reading every frame; clips under 10 s still give one frame; raise memory and timeout, since Cloud Run's /tmp uses memory |
| 8. Upload page | Several files upload with live progress; duplicates show a clear already-uploaded message | `fetch` reports no upload progress, so PUT with `XMLHttpRequest`; hashing a large video in the browser holds the whole file in memory |
| 9. Delete API | A list of URLs removes originals, thumbnails, frames and records, with a result per URL | Delete objects first and the record last, ignoring not-found, so a retry finishes a half-done delete; share one URL helper with member 4's bulk tag API |
| 10. Delete in gallery | Selecting tiles and confirming removes them from storage and the gallery | Build against a plain list first if the gallery isn't ready |
| 11. Roles and edge tests | Each function runs under its own service account with only what it touches, and edge cases pass | Test the same file twice, the same file renamed, each image format, a clip under 10 s, deleting a video, deleting twice, and no token |
| 12. Demo uploads, report parts | Demo files went through the real pipeline; your user guide steps are written | Upload demo data before the Saturday freeze so tags and thumbnails are real, not seeded |

## Dependencies and conflicts

Your biggest exposure is to member 3, who shares your Firestore record and your video frames. Member 1's project access and gateway are the first things you wait on.

| Member | Direction | What | Needed by | If it slips or clashes |
|---|---|---|---|---|
| 1 | You need | Shared GCP project and AWS account access | Oct 26, 2026 | Nothing deploys. Build in your own GCP project meanwhile, with project and bucket names in a config file so switching is one edit |
| 3 | Shared | Frame handoff (in memory or through the derived bucket), with counts as the per-frame maximum | Oct 26, 2026 | You and member 3 build halves of video tagging that don't fit together |
| 1, 3, 4 | They need | Schema, key naming and route contracts | Oct 26, 2026 | Teammates guess field names, and queries, tagging and triggers break at integration |
| 1 | You need | Gateway routes for init and delete that pass the user's claims through | Oct 27, 2026 | Test against your functions' direct URLs with a fixed test user until the gateway is ready |
| 3, 4 | They need | Seed data script | Oct 27, 2026 | Member 4 can't build queries, and member 3 can't test lost pet search, until real tags exist |
| 3 | They need | First real upload creating a record (Gate 2) | Oct 27, 2026 | Member 3 can't connect tagging to uploads, so the whole Pipeline phase slips |
| 3 | Shared | Both functions write the same Firestore record | Oct 28, 2026 | A full overwrite by either function wipes the other's fields; both use merge writes only |

| Member | Direction | What | Needed by | If it slips or clashes |
|---|---|---|---|---|
| 3 | Shared | Query photos never reach the media bucket | Oct 28, 2026 | Query-by-file photos get stored and tagged like uploads, breaking the spec's privacy rule |
| 4 | They need | Thumbnail keys on records | Oct 29, 2026 | The gallery shows broken images and the thumbnail lookup fails |
| 3 | They need | Video frames | Oct 29, 2026 | Videos never get tagged |
| 1 | You need | App shell and shared API client | Oct 29, 2026 | Build the upload page standalone and move it into the shell on Saturday |
| 1 | You need | Notification trigger that ignores deletes and non-tag writes | Oct 30, 2026 | Every upload, thumbnail write or delete emails subscribers |
| 4 | Shared | One URL-to-file-ID helper for delete and bulk tagging | Oct 30, 2026 | One API accepts presigned URLs and the other returns 404. Write it yourself, since you own the naming |
| 4 | You need | Gallery with tile selection | Oct 30, 2026 | Ship delete as a plain list view and plug it into the gallery later |
| 1 | You need | Review of your service accounts | Oct 31, 2026 | Over-broad roles get exposed by the examiners' permissions questions in the demo |